Bid intelligence for Weibee Steel: the full walk-through

Where we win, what the work actually earns, and which of the next two years' opportunities deserve estimating hours. 413 bids, 36 general contractors, 90 upcoming projects.

Made from synthetic data. Every number in this deck comes from a generated bid history built to behave like a mid-size fabricator; none of it is Weibee Steel's data. The method runs unchanged on real history.

Deepak Reddy | Business Development Associate candidate

What this deck covers

1. **The problem and the approach**
   What we cannot answer today, the four-step loop, the data it needs
2. **What the data says**
   Hit rates, pricing feedback, customers, estimate versus actual, shop capacity
3. **The decision tool**
   How the scorecard scores a project, a worked example, the ranked list, the rules
4. **How much to trust it**
   What drives the win odds and how the model tested out of sample
5. **Making it real**
   From the sample to Weibee data, a 90-day plan, technology, risks, the monthly KPIs

1. The problem and the approach

Every bid teaches us something. Today the lesson stays in people's heads.

Four questions a fabricator should be able to answer, and usually cannot



How often do we win, by what?

Hit rate by project type, size, distance, customer and how the job is bought, with a denominator. Not a feeling about schools.



What did the work actually earn?

Estimated versus realised gross margin by kind of job, so the next estimate learns from the last close-out.



Why did we lose?

Price gap to the low number when we lost on price; relationship, schedule or scope when we did not. Collected every time.



Is this invitation worth a week?

A ranked list of upcoming work, scored on odds, margin, estimating cost and shop load, before the takeoff starts.

The loop: collect, measure, rank, decide



1. Collect

Five tables: customers, opportunities (bid or not), bids, jobs, upcoming pipeline. Plus the shop calendar.

Produces: one source of truth



2. Measure

Hit rate by count and by dollars. Won dollars per estimating hour. Realised margin by kind of work. Price gap on losses.

Produces: the report's charts



3. Rank

Win odds from our own history. Expected margin dollars net of estimating cost. Shop load if we win it.

Produces: the scorecard



4. Decide

Monday bid meeting works the ranked list. The room overrides, the override is logged and becomes training data.

Produces: better odds next quarter

The data it needs: five tables, most of which already exist somewhere

Table	One row per	Core fields	Where it lives today
Customers	General contractor or owner	Name, HQ, tier, first contact	Estimators' contacts, QuickBooks
Opportunities	Project we heard about, bid or not	GC, type, city, miles, tons, complexity, delivery method, scope, due date, fab window, bid / no-bid and why	Email, the whiteboard
Bids	Number we submitted	Estimating hours, cost build-up, markup, price, bidders, result, lost reason, low bid number	Estimating log, bid tabs
Jobs	Won bid, filled in as it runs	Actual hours and material, change orders requested and approved, final revenue, schedule variance, NCRs, days to pay	Job cost, PM close-outs
Pipeline	Project expected to bid in 6 to 24 months	Same as opportunities plus status: lead, invited, bidding	Does not exist yet

The two fields that are missing almost everywhere: the no-bid decision with its reason, and the low number when we lose on price. Coverage of those two fields is the first KPI I would report, because every other number depends on them.

2. What the data says

413 bids over four and three-quarter years, 95 completed jobs, 36 general contractors. Shaped like a real shop's history.

The business at a glance, trailing twelve months

25%

Hit rate

20 of 80 decided bids; prior year 33%

$19.4M

Value won

5,088 tons

16 weeks

Won work waiting for the shop

2,286 tons at 140 tons a week

50%

How full the shop looks next year

won work plus upcoming work at its odds

4.5 pts

Margin estimated but not kept

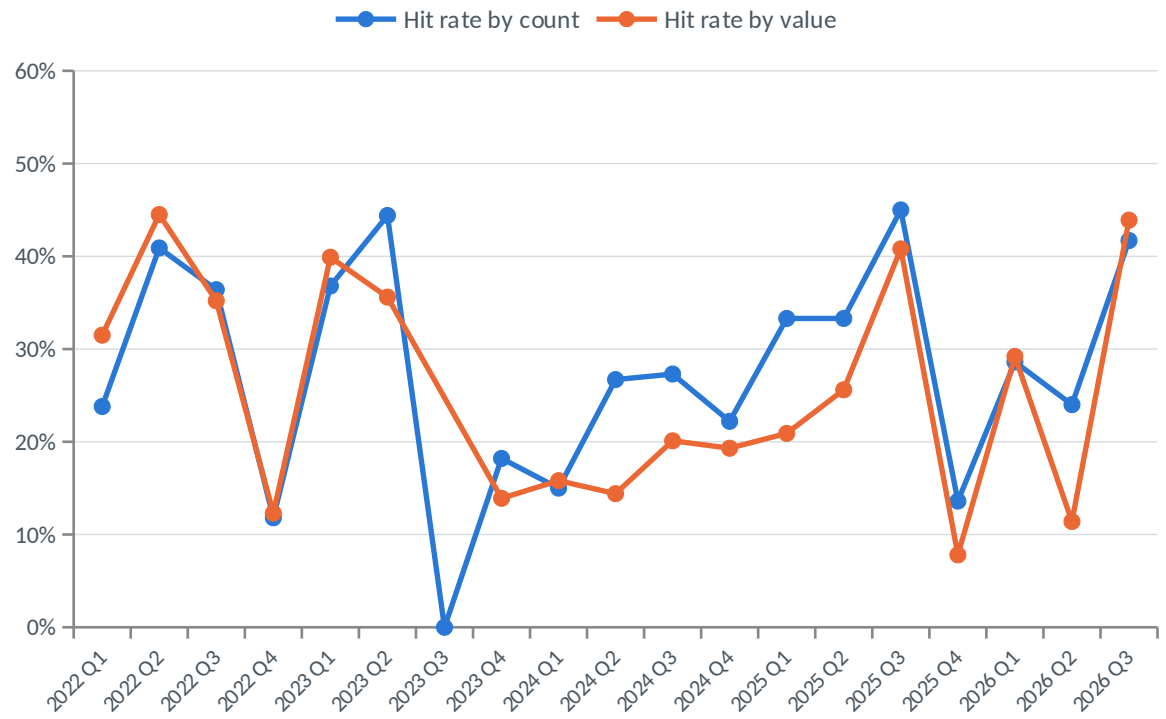
estimated 22.9%, actually earned 18.3%

29 of 90

Upcoming projects worth chasing

about 698 estimating hours to bid them

Hit rate by quarter: by count and by dollars



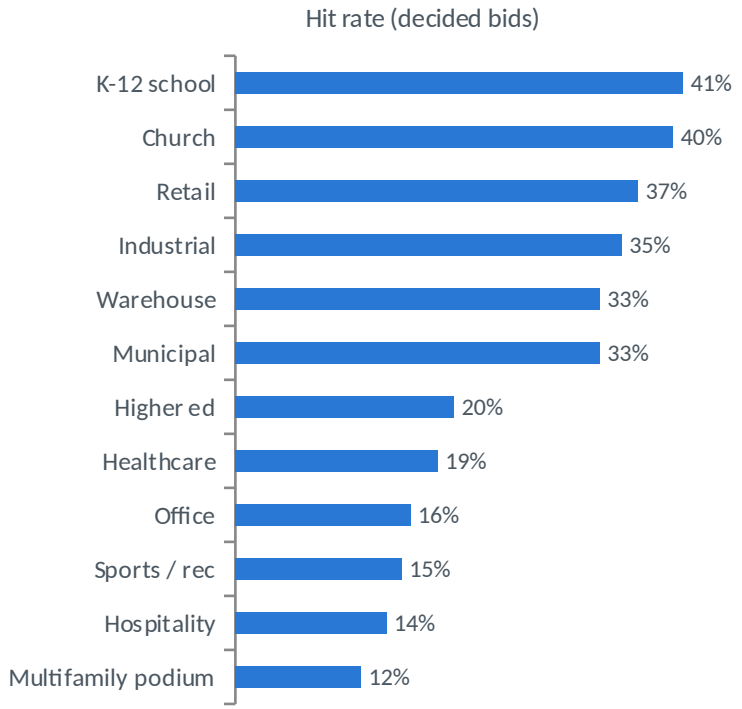
Decided bids only; cancelled projects do not count against us.

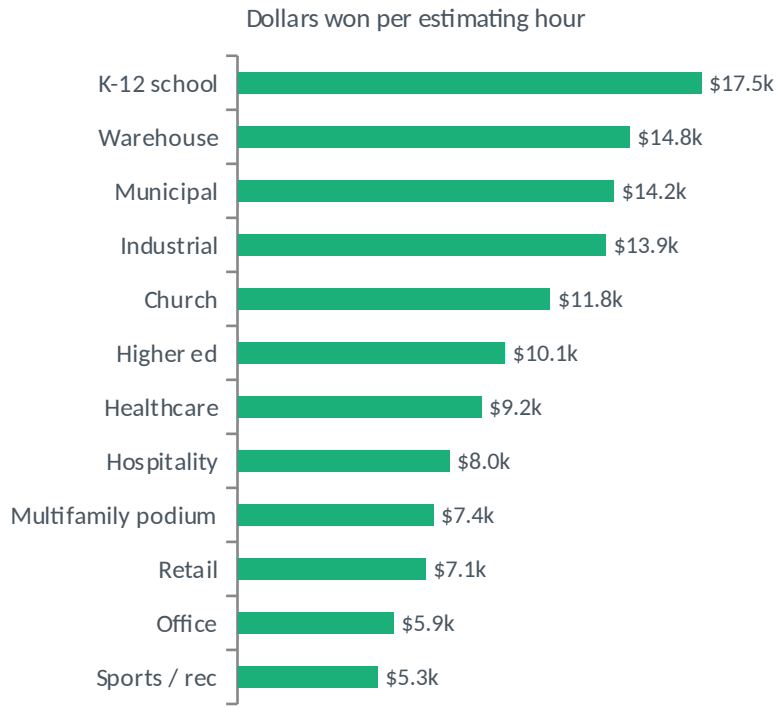
Long-run average 27%. Quarters swing because 20 bids is a small sample; one quarter is weather, four is a trend.

Count and dollar rates move together, so we are not winning only the small jobs.

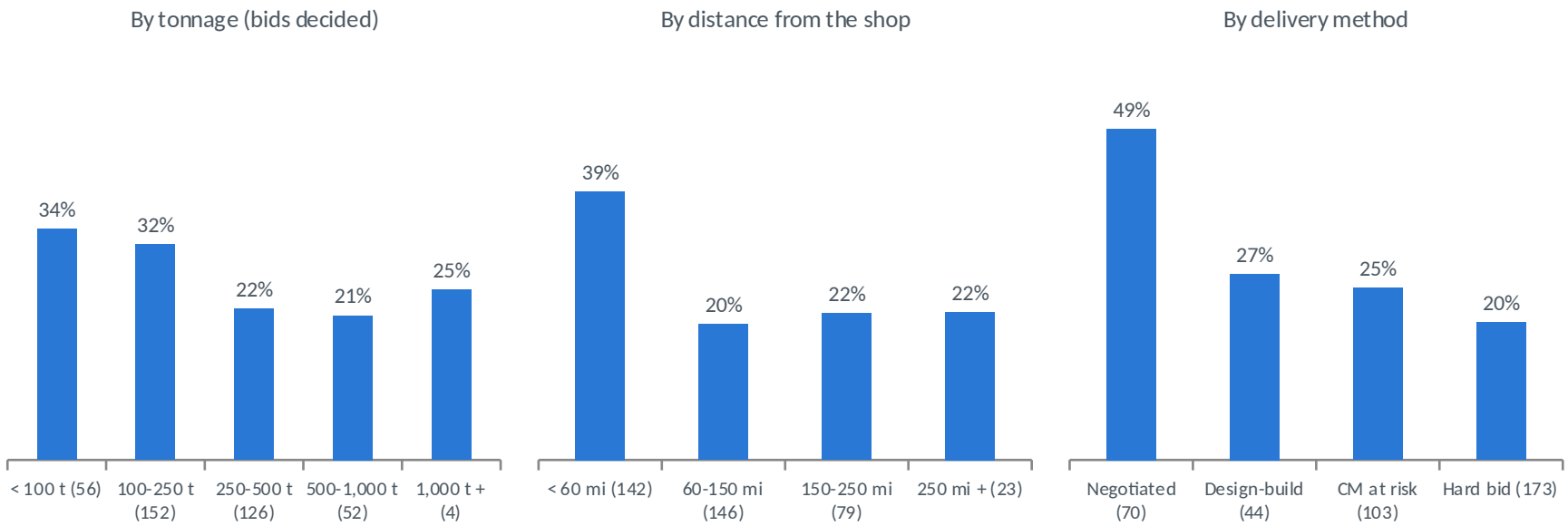
Report both every quarter and never react to one.

Where the shop fits: hit rate and dollars won per estimating hour, by project type



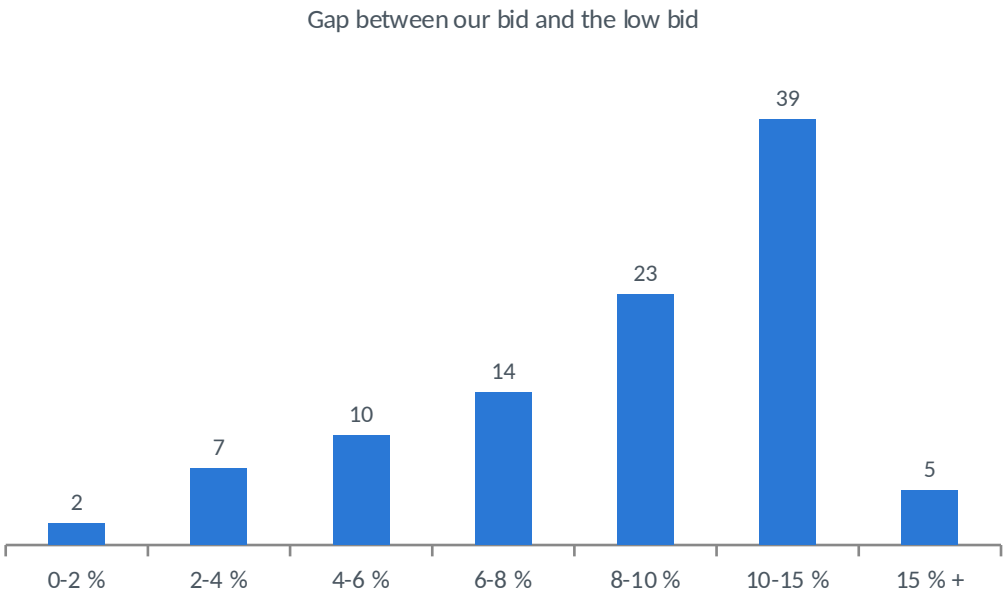


Hit rate by size, distance and how the job is bought



Reading it: negotiated and design-build work wins roughly twice as often as hard bids; hit rate falls off beyond 250 miles (23 bids); and the mid-size bands carry most of the dollars. None of these cuts is a reason to never bid, all of them are a reason to bid differently.

Pricing feedback: how far above the low number were we when we lost on price?



9.7%

Median gap when we lose on price

feedback collected on 62% of price losses

3.9%

Left on the table when we win

median gap to the next-lowest bid, where known

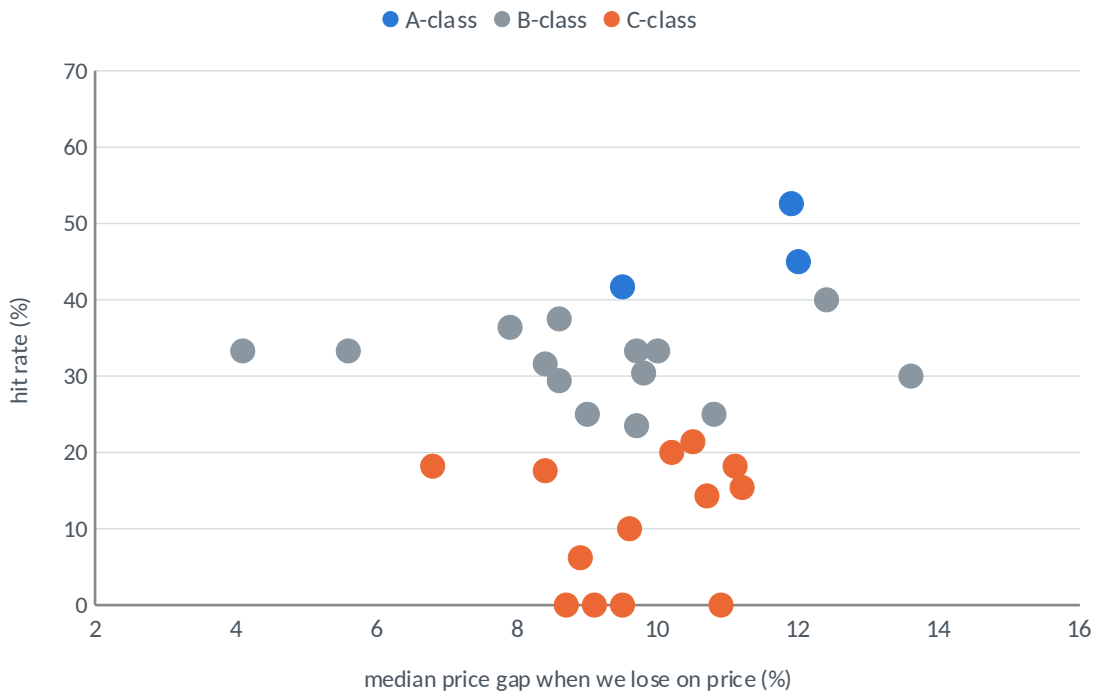
Rule of thumb: a gap under 3% is a markup conversation. A gap over 10% means we should not have been on that bid list, and the scorecard should learn that. Most of our losses sit in the second group.

Customer scorecard: the ten general contractors we have won the most from

General contractor	Tier	Class	Bids	Hit rate	Won	Won / est. hr	Price gap	Realised margin	Days to pay
Sooner State Contractors	regional	A	20	45%	$11.8M	$18k	12.0%	17.2%	62
Granite Peak Builders	regional	A	24	42%	$8.4M	$14.2k	9.5%	20.5%	58
Wagon Trail Builders	national	A	19	53%	$8M	$16.4k	11.9%	22.6%	47
Ouachita Building Group	regional	B	17	24%	$6.8M	$13.1k	9.7%	15.4%	49
Meridian Build Partners	regional	B	23	30%	$6.5M	$12.2k	9.8%	17.7%	61
Washita Valley Builders	local	C	17	18%	$6.3M	$15.4k	8.4%	15.8%	63
Longhorn Commercial	regional	B	9	33%	$6.1M	$21.4k	5.6%	28.3%	47
Great Plains Constructors	national	B	18	33%	$5.6M	$12.1k	9.7%	22.6%	71
Red River Builders	regional	B	19	32%	$5.5M	$12.3k	8.4%	20.1%	42
Verdigris Construction	local	B	9	56%	$4.8M	$19.5k		15.0%	62

A = awards us 40% or more and the work earns at or above shop margin. B = 22% or more. C = below. Hit rate alone flatters a GC who sends us everything; realised margin, change-order behaviour and payment speed belong next to it.

Who awards us work and who checks their numbers against us



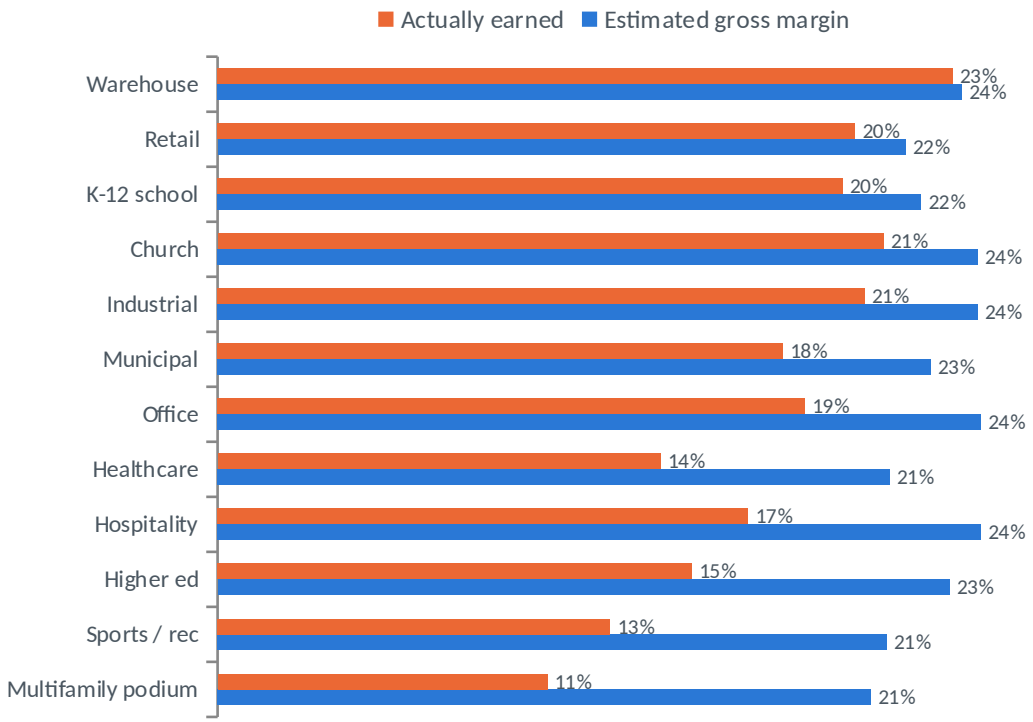
Bottom right is the expensive corner: GCs who rarely award us and are far below us on price.

Top left is where the business is: GCs who award us often and whose price gap is small when they do not.

Each dot is a GC with three or more decided bids.

Spend estimating hours top-left, and on new GCs that look like them.

What the estimate promised versus what the job earned, by project type



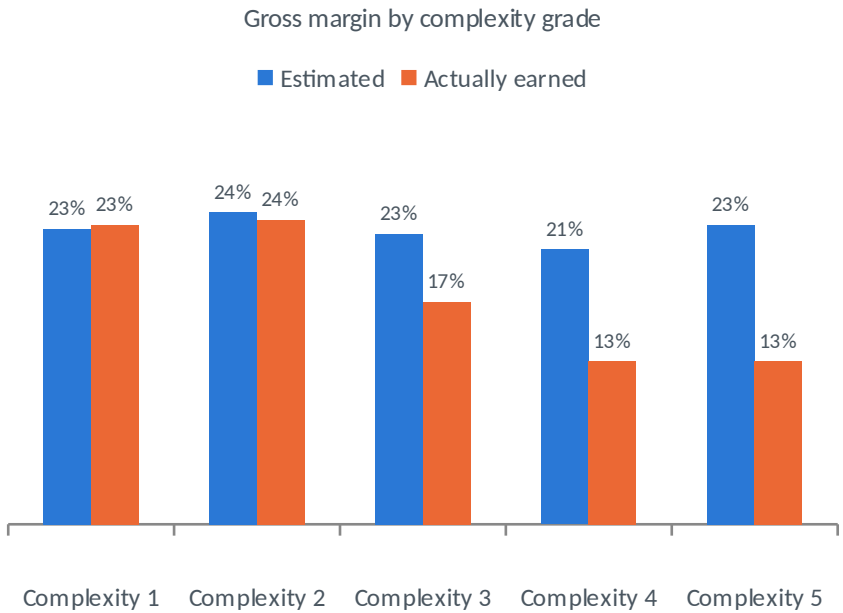
Across 95 completed jobs: 22.9% estimated, 18.3% earned.

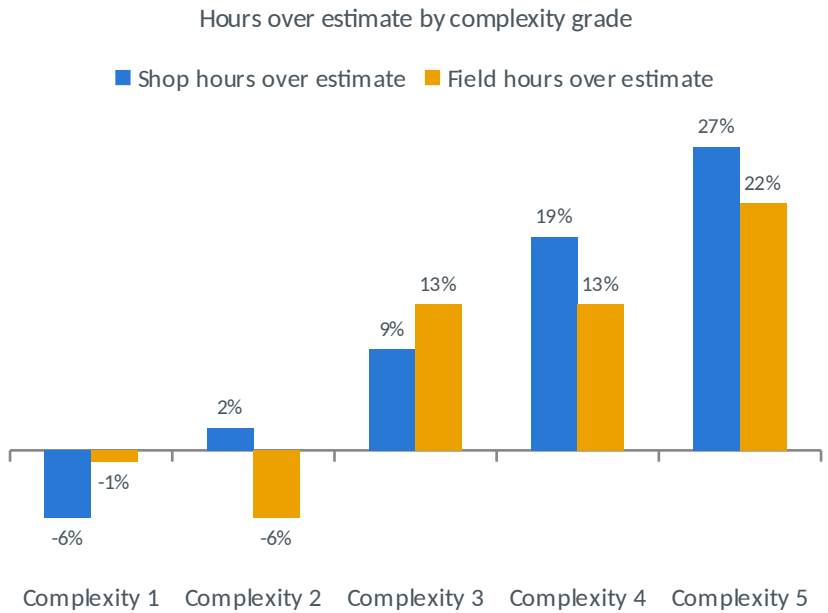
4.5 points, about $4.3M, that was in the bid and not in the bank.

Sorted worst to best. The estimate is roughly right on simple work and roughly four to ten points optimistic on complex work.

Put realised margin by type into the estimate as a line item.

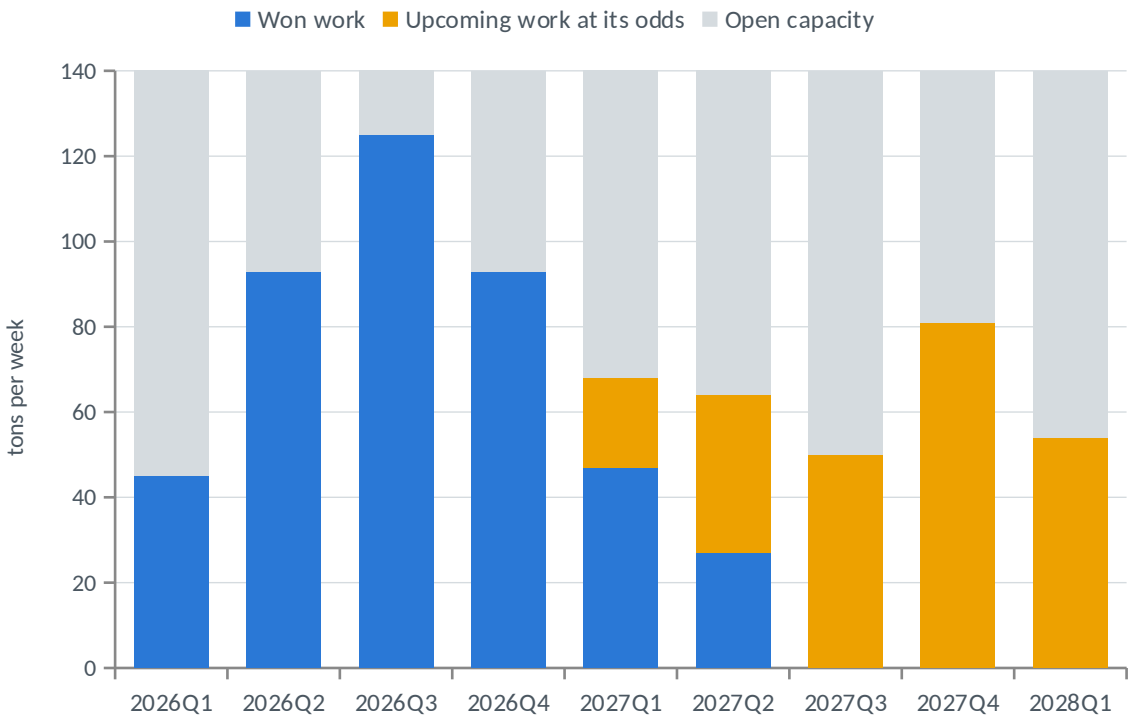
Complexity is where the margin goes: hour overruns by complexity grade





Grades: 1 is simple braced frames, 5 is architecturally exposed steel, long spans or heavy moment connections. Shop hours ran 10% over estimate overall and field hours 8%; the overrun grows with the grade. A complexity-based contingency, not a flat markup, is the fix.

Can the shop take what we are chasing? Backlog and weighted pipeline against capacity



Each won job spreads its tonnage evenly over its fabrication weeks.

Each upcoming job counts at tons times win odds: a 400-ton job at 25% is 100 tons.

Won work covers 16 weeks. From early next year the shop is less than half full unless the pipeline converts.

Target: roughly 9,000 more tons of identified opportunities at today's hit rate.

3. The decision tool

Every upcoming project scored before a week of takeoff starts. A starting point for the bid meeting, not a verdict.

How the scorecard scores a project

Win odds

From a model fitted to our own bids: project type, tonnage, miles, complexity, delivery method, bidders, and our history with that GC.

Expected revenue

Tons times our historical price per ton for that project type (the real estimate replaces this once one exists).

Margin we really keep

Realised gross margin on that project type from completed jobs, not the estimate's margin.

Estimating cost

Hours from a curve fitted to our own bid log (tons and complexity) times the loaded estimator rate.

Expected value = win odds x expected revenue x margin we really keep minus estimating cost

Ranked per estimating hour, because hours are the scarce input, and compared with what an hour has returned historically (about $2,468 of gross margin per estimating hour over the last two years). Pursue at 80% of that or better, Review from 35% to 80%, Pass below. A separate check projects shop load over the job's fabrication weeks if we win it.

Worked example: one Pursue and one Pass, step by step

Pursue Industrial - Stillwater #24

Caddo Construction, Stillwater, 83 mi, 398 t, complexity 1, Design-build, 3 prior bids with this GC at 14% hit rate

Step	Value
Win odds	52%
Expected revenue	398 t x $4,011 = $1.60M
Margin we really keep	19.9%
Expected margin dollars	52% x $1.60M x 19.9% = $167k
Estimating cost	25.8 h x $85 = $2k
Expected value	$164k
EV per estimating hour	$6,373 (258% of historical)
Shop load if won	82%
Reasons	design-build delivery

Pass Warehouse - Fort Smith #83

Longhorn Commercial, Fort Smith, 195 mi, 711 t, complexity 1, Hard bid, 9 prior bids with this GC at 32% hit rate

Step	Value
Win odds	11%
Expected revenue	711 t x $3,130 = $2.23M
Margin we really keep	21.9%
Expected margin dollars	11% x $2.23M x 21.9% = $51k
Estimating cost	37.4 h x $85 = $3k
Expected value	$48k
EV per estimating hour	$1,283 (52% of historical)
Shop load if won	79%
Reasons	win odds 11%

The ranked pipeline: 90 upcoming projects on one chart



Pursue — **29 projects**

698 estimating hours
9,872 tons, $2.5M expected margin net of estimating

Review — **26 projects**

451 estimating hours
4,608 tons, $0.72M expected margin net of estimating

Pass — **35 projects**

858 estimating hours
9,961 tons, $0.53M expected margin net of estimating

The top ten, as the bid meeting would see them

#	Call	Project	GC	Tons	Miles	Delivery	Odds	EV / hr	Shop load	Why
1	Pursue	Industrial - Stillwater #24	Caddo Construction	398	83	Design-build	52%	$6.4k	82%	design-build delivery
2	Pursue	Municipal - Norman #23	Flint Hills Construction	134	13	Negotiated	83%	$5.6k	74%	negotiated delivery
3	Pursue	Industrial - Muskogee #3	Meridian Build Partners	494	133	Design-build	43%	$5.2k	72%	design-build delivery
4	Pursue	Industrial - Oklahoma City #11	Boomer Construction Group	923	32	CM at risk	36%	$5k	69%	no flags
5	Pursue	K-12 school - Tulsa #26	Sooner State Contractors	175	120	Negotiated	65%	$4.7k	63%	relationship GC (43% hit rate, 20 bids); project type we w…
6	Pursue	Industrial - Oklahoma City #45	Summit Ridge Construction	1330	30	Hard bid	24%	$4.2k	69%	no flags
7	Pursue	Municipal - Tulsa #61	Meridian Build Partners	196	126	Design-build	53%	$4.1k	62%	design-build delivery
8	Review	Municipal - Muskogee #32	Cimarron Commercial	145	136	Negotiated	62%	$3.9k	98%	shop tight (98%) during fabrication window; negotiated del…
9	Pursue	Industrial - Stillwater #49	Flint Hills Construction	338	83	CM at risk	37%	$3.9k	55%	no flags
10	Pursue	Industrial - Moore #30	Washita Valley Builders	414	21	CM at risk	36%	$3.7k	75%	no flags

Full list of 90 is in the report and the CSV, with every input next to every call so the room can argue with the number.

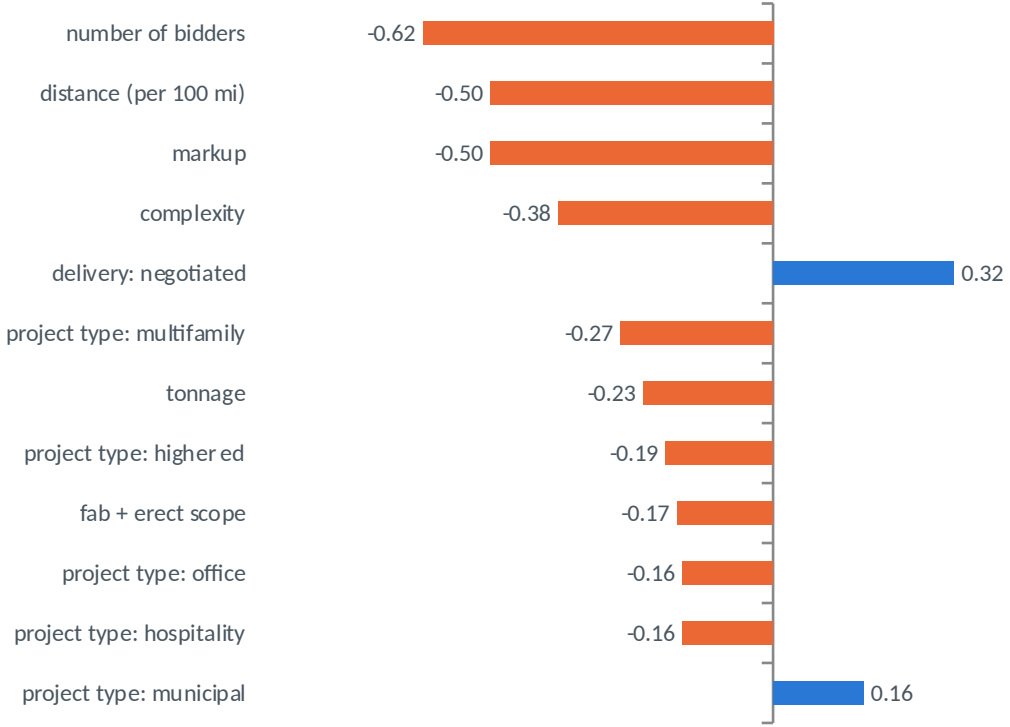
The rules, in the order they fire, and how the room overrides them

1	**Hard flags first**	Shop over 105% of capacity if we win it, or win odds under 12%, is a Pass whatever the dollars say.
2	**Then the money**	Expected value per estimating hour against our historical return: 80% or better is Pursue, 35% to 80% is Review, below is Pass.
3	**Then the soft flags**	Shop tight (over 90%), erection beyond 250 miles, a project type where we realise well below shop margin: a would-be Pursue becomes Review.
4	**One strategic override**	A new regional or national GC on one of our three best-fit project types is never an automatic Pass; it is Review, labelled strategic.
5	**The room decides**	Every call shows its inputs and reasons. When the room overrides, the override and the reason are logged and become next quarter's training data.

4. How much to trust the win odds

A small statistical model on our own bids, tested on bids it had never seen. Good enough to rank, not good enough to overrule the room on one job.

What moves the win odds, in order of effect



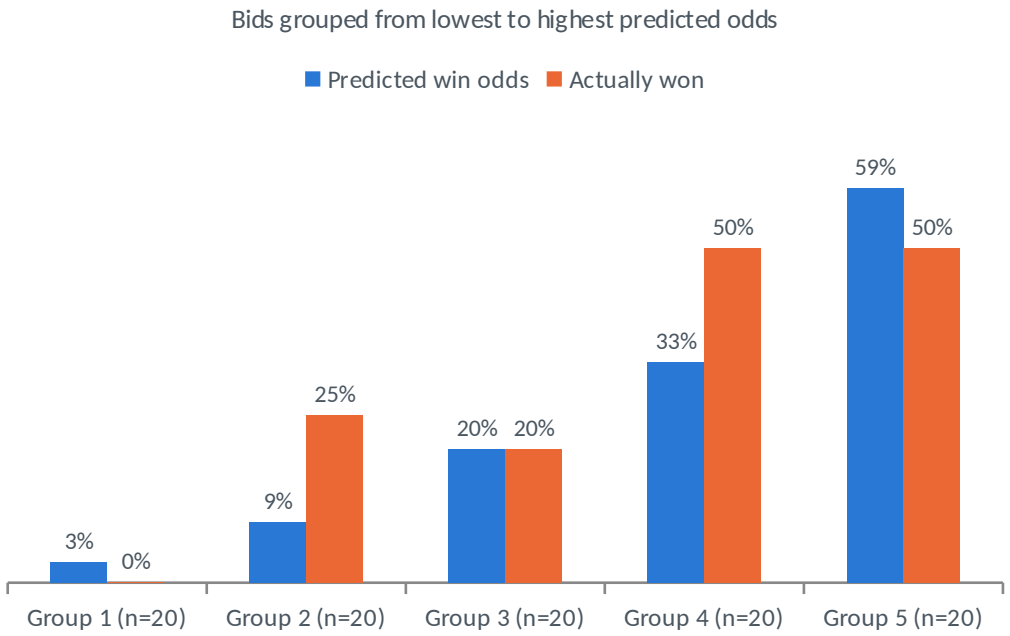
Bars to the left lower our odds, bars to the right raise them.

More bidders, more miles, a higher markup and higher complexity all cost us. Negotiated delivery and a good history with the GC help.

Measured against warehouse work on a hard bid.

Nothing here surprises an estimator. That is the point: the model agrees with experience and adds a number.

Did the odds come true? Tested on bids the model had never seen



0.75

Accuracy score (AUC)
0.5 is a coin flip, 1.0 is perfect

0.182 vs 0.206

Error score vs guessing the average
Brier score, lower is better

100 bids

Held out for the test
fitted on 290 bids before 2025-07-01

How to use it: the groups line up, so a 40% call wins about 40% of the time. Use the odds to rank and to budget hours. Do not use them to skip a job the room believes in. Re-fit every quarter; every decided bid and every low-bid number makes it better.

5. Making it real at Weibee Steel

What changes with real data, the first 90 days, the technology worth evaluating, the risks, and what we would report every month.

From the sample to Weibee's own numbers

Piece	In this sample	On real data
Source data	Sample history shaped like a mid-size fabricator	Estimating log, job-cost close-outs, GC list, loaded through the same five tables
Capacity	140 tons a week, one step change	The shop calendar with shifts, holidays and outsourced fabrication
Expected revenue	Historical price per ton by project type	The actual estimate once one exists; type median only at lead stage
Win model	Type, size, miles, complexity, delivery, bidders, GC history	Add the estimator, bid-tab bidder names, time since last award, contact changes
Thresholds	Relative to our historical return per estimating hour	Same; reviewed quarterly with the estimating manager
The report	One HTML page, a PDF, a Tableau data layer	Same files on the shared drive, or the same tables behind Power BI or Tableau
Findings	Illustrate the kind of result	Become decisions with names and dates attached

The first 90 days

Weeks 1 to 4

Learn the shop, build the intake

- Shadow estimating, PM, QC, shop and field as the posting describes
- Stand up the five tables; load three years of bid logs and job closes
- Phone the GCs we lost to for the low number; log no-bids going forward
- KPI: coverage of lost reason and low-bid fields

Weeks 5 to 8

Get the numbers live

- Hit rate, customer and estimate-versus-actual pages on real history
- Sit with estimators and PMs until definitions are agreed: decided bid, direct cost, change-order booking
- First account plan from the customer scorecard
- KPI: the report runs weekly without me touching it

Weeks 9 to 13

Turn on the Monday scorecard

- Score the pipeline; set the quarterly estimating-hour budget
- Log every override with a reason
- Monthly capacity review with shop leadership
- KPI: share of estimating hours on Pursue work; pipeline tons versus the gap

Technology worth evaluating, in this order, and how we would judge each



1. GC and opportunity database

A spreadsheet with discipline first, then a light CRM. Judged on whether the intake survives staff changes.



2. Automated and AI-assisted takeoff

Model-based quantities from Revit or IFC, or AI-assisted PDF takeoff. Judged against our fitted hours-per-bid curve on the project types we actually pursue.



3. Lead services

Dodge, ConstructConnect, BuildCentral. Every lead runs through the scorecard, so we buy leads where the model says we win.



4. Pricing feedback

A post-award email template and a field in the database before any software. Cheapest, highest return.

Risks and limits, stated up front

Garbage in

If lost reasons and low-bid numbers are not collected, the pricing section and the odds degrade quietly. Mitigation: coverage is a reported KPI.

Small samples

A few hundred bids. Quarterly hit rates swing, project-type effects are noisy. Mitigation: report sample sizes, shrink thin cuts toward the average.

False precision

A 38% win probability is not a fact. Mitigation: show odds as ranges in the room, never overrule a person on one job.

Gaming

If hours or markup are judged by the scorecard, people will manage to it. Mitigation: the scorecard allocates hours; it does not grade estimators.

Scope

Only estimating hours are modelled as scarce. Bonding, cash, crews and GC schedule risk belong in the reasons column once data exists.

Adoption

If the Monday meeting does not work the list, it is a report nobody reads. Mitigation: the estimating manager owns the rules and the hour budget, not the analyst.

What we would report every month

KPI	Definition	Why it matters
Hit rate, count and value	Wins over decided bids; dollars won over dollars bid, trailing 12 months	The headline, both ways, so we see if we are winning only the small jobs
Won dollars per estimating hour	Value won over all estimating hours spent, by project type	The return on the estimating team's time
Feedback coverage	Share of price losses with the low number recorded; share of no-bids with a reason	Everything else depends on it
Median price gap	Our bid over the low bid when we lost on price, by delivery method	Markup problem or fit problem
Margin slip	Estimated minus realised gross margin on jobs closed this quarter, by type and complexity	Whether the estimate is learning
Pipeline coverage	Backlog plus weighted pipeline over capacity, next four quarters	The business-development target in tons
Hours on Pursue work	Share of estimating hours spent on Pursue calls; override count and reasons	Whether the scorecard is being used, and where it is wrong
A-class share	Share of bids and of won dollars from A-class accounts; new A-class candidates	Account planning

Every bid teaches us something. This puts the lesson in a table and on the Monday agenda.

Report, Tableau data, technical design and code are in the repository. The method runs unchanged on real history.

Thank you